

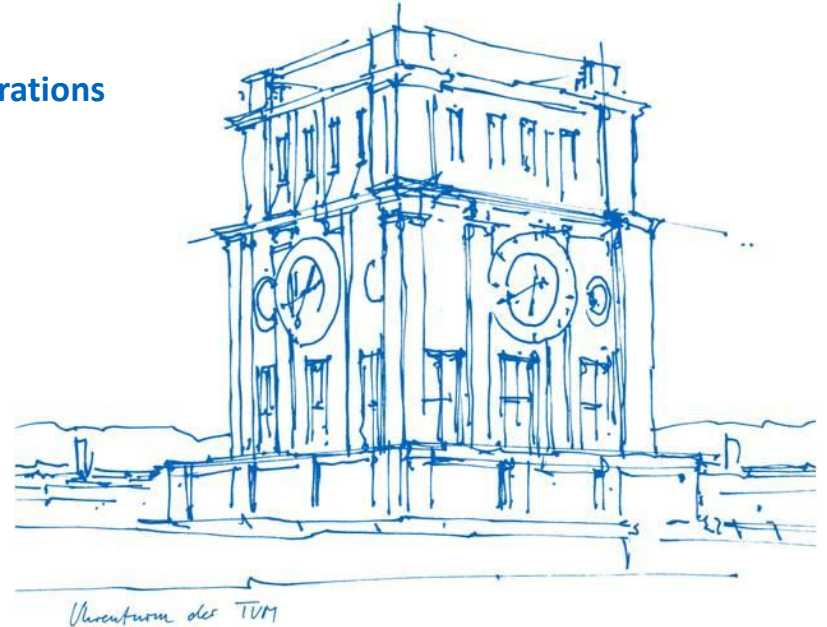
Pressing Topics in Next Generation Data Centers

**Advanced Practical Course – Next Generation Data Center Operations
for SAP Enterprise Software (IN2128, IN2106, IN212810)**

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22 April, 2026



Agenda

I. AI & GPU

II. Cloud & On-Premise

III. Access Control

IV. Cybersecurity

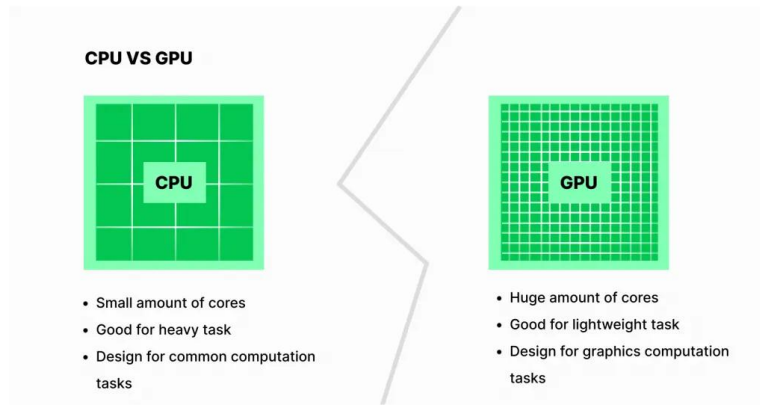


Source: <https://www.nexdatacenter.com/get-prepared-for-the-features-of-next-generation-data-center-technologies/>

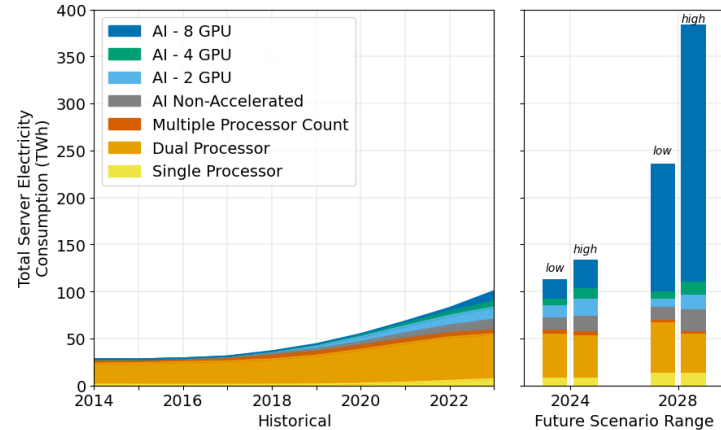
AI & GPU

Hardware-Hungry AI

The rise of AI has changed the paradigms of computing center operations fundamentally. GPUs become the working horses of AI-dedicated computing centers.



Source: <https://medium.com/@reyes83/cpu-vs-gpu-for-model-training-understanding-the-differences-a82c8d016293>



Source: <https://iee.psu.edu/news/blog/why-ai-uses-so-much-energy-and-what-we-can-do-about-it>

Currently, GPU Power is far more expensive than CPU Power

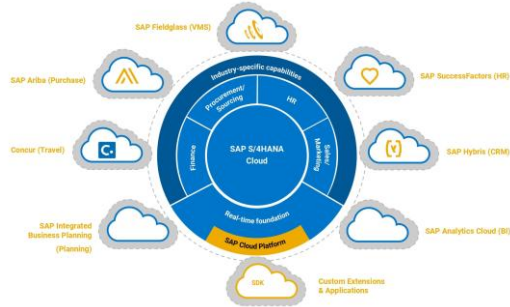
AI triggers a Paradigm Shift in Data Center Architecture

- AI has different kinds of load
 - Training: Long-time, resource intensive, planable
 - Inference: Short-time, less resource intensive, unplanable
- GPUs have far greater energy density
 - 72 CPU Cores (Power 10) → 3 kW
 - 4 GPUs (H100) → 4 kW
 - Servers with more than 4 GPUs need water-cooling
- Server rooms are far more limited by cooling capacity and power supply than physical space
 - Many present server rooms have not been built with GPUs in mind
- GPU needs other types of virtualization
 - more power condensed in less units
 - Classical sharing concepts might only work in very large clusters
 - LLM training load often needs multiple GPUs at work
- GPUs often not feasible for on-premise installation
 - Data Centers must be hybrid → Hardware for inference local, while Training Hardware is in the cloud

Cloud vs. On-Premise

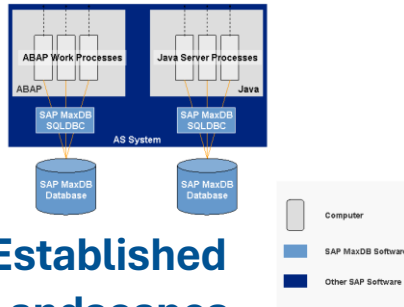
A Climate of External Demands in Tension

Source: <https://www.sileron.com/en/sap-s4hana-cloud-erp/>



Push to Cloud

Source: https://maxdb.sap.com/doc/7_7/44/fc2e5d1d3c6c30e10000000a114a6b/content.htm

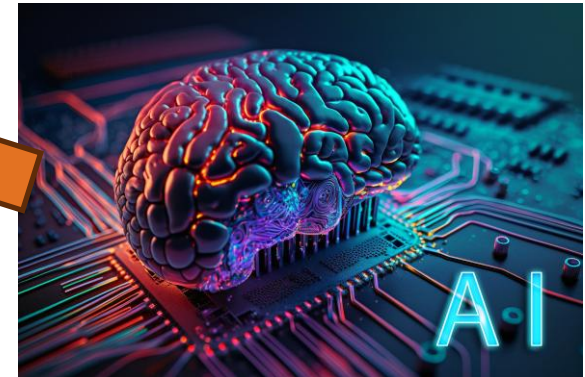


Established Landscapes

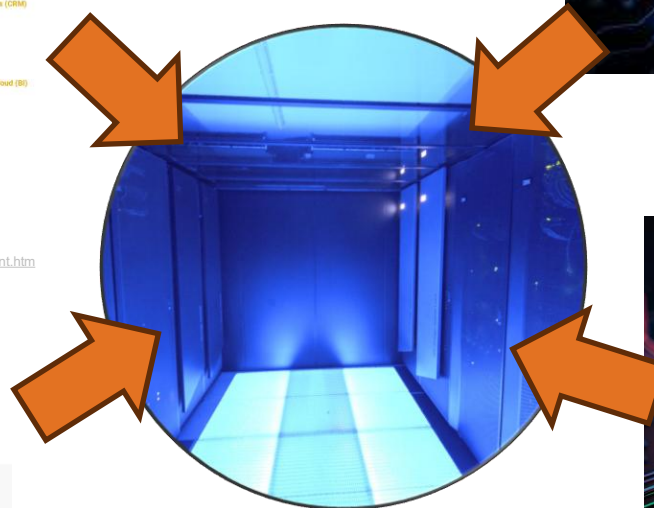


Security Concerns

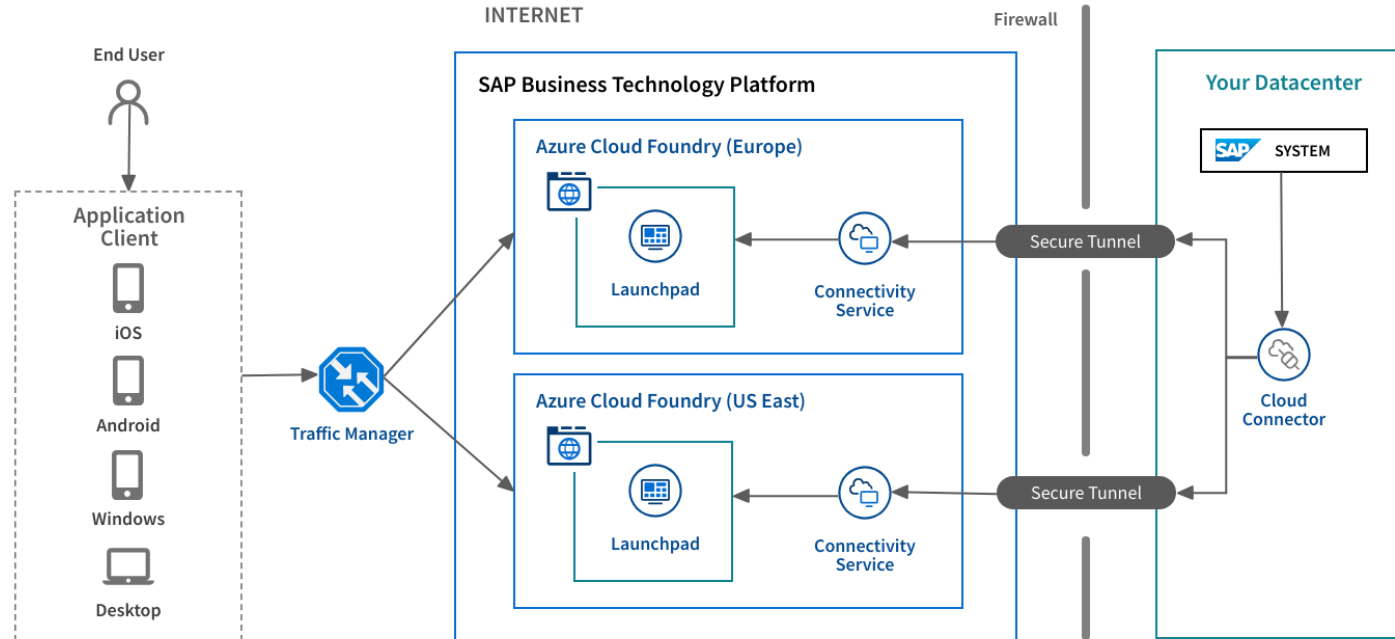
Source: <https://effvision.com/how-is-ai-getting-used-nowadays/>



Hosting AI



Hybrid Data Center Landscapes



Source: <https://mockflow.com/flowchart-examples/hybrid-cloud-architecture-diagram-example-1>

Access Control

Role-Based Access Control

Flat RBAC Model (See [Sandhu et al., 2000](#))



Flat RBAC Model (See [Sandhu et al., 2000](#))



Trade-Off:

- Usability / Performance:
- Security / Principle of Least Privilege (See [Saltzer & Schroeder, 1975](#)):

Flat RBAC Model (See [Sandhu et al., 2000](#))

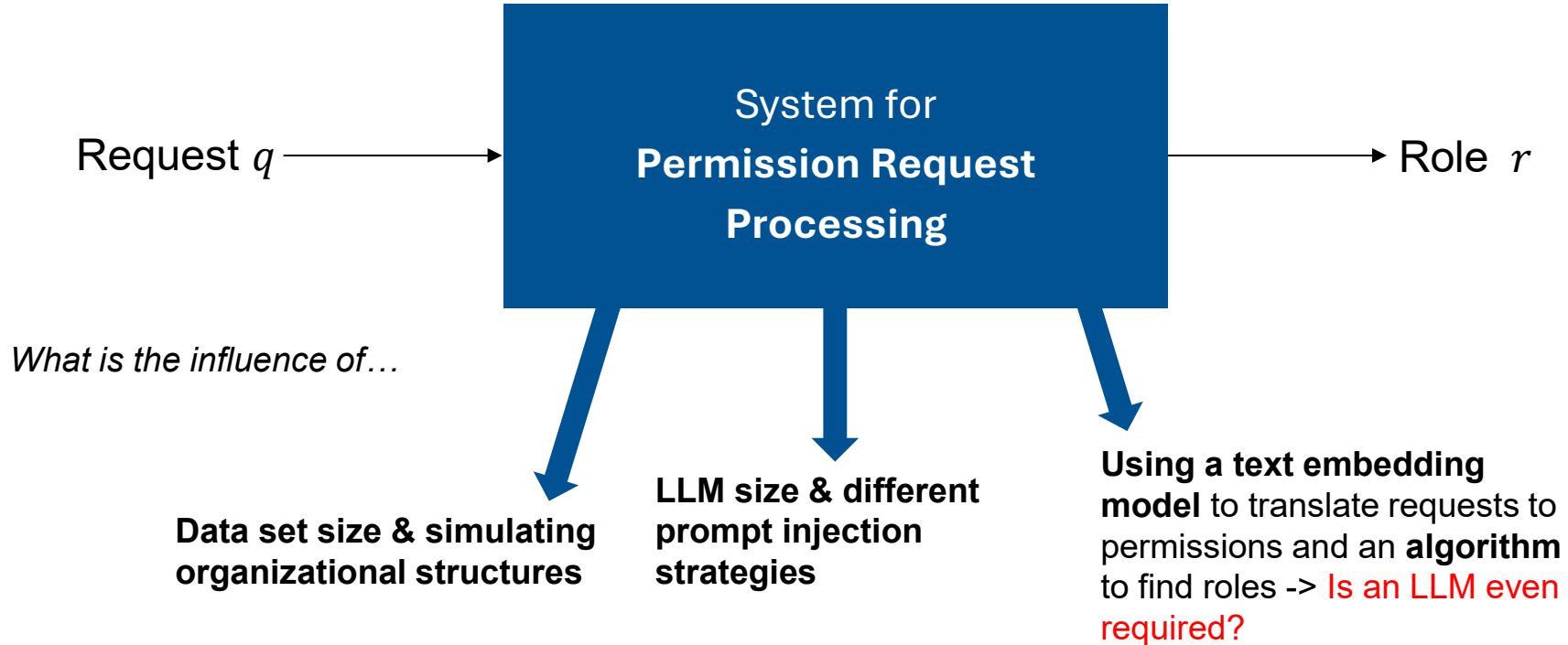


Trade-Off:

- Usability / Performance: Everyone has admin rights! And can do *everything*.
- Security / Principle of Least Privilege (See [Saltzer & Schroeder, 1975](#)): Every user does their daily jobs with the minimum required permissions assigned. -> **Why is this not usable?**
 - Each time a user requests some new permission $p \in P$ -> you need to search for the correct role $r \in R$ s.t. $(p, r) \in PA$
 - **A request is in human language** -> "I need to do something." and not "I need permission XYZ."

Permission Request Processing

Parameters



On Hosting

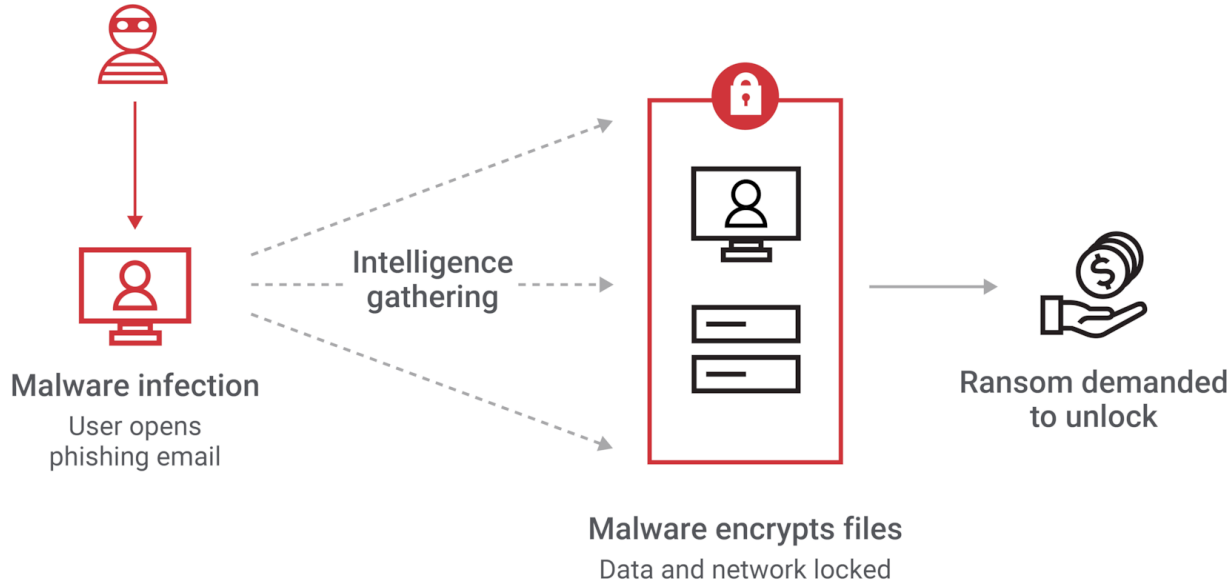
- We already found that the capability tier of an LLM is not that important, but rather that an LLM knows the enterprise data (see [Niedermeier et al., 2025](#))
 - -> You can not simply use some online LLM, but in practice, you must use a self-hosted one.
- Token Per Second (TPS) and Time To First Token (TTFT) are not that important factors in this use case, as there are (hopefully) not hundreds of permission requests per day. If so, you have another problem...
- Only delegate tasks to the LLM where you really need it.
 - Example: If the LLM gives me the correct permission and I fully know the RBAC model, then I can just run a deterministic algorithm that provides me with the smallest role that covers that permission. -> No need to query the LLM a second time.

On Transparent LLM Research

- Accuracy: Does the LLM suggest correct permissions and roles?
- Consistency: Does the LLM suggest the same permissions and roles when sending the same request multiple times? (See [Song et al., 2025](#))
- Hallucination: Does the LLM hallucinate contents? (E.g., see [Perković et al., 2024](#))
- Modesty: Does the LLM admit to not know the correct answer instead of providing a wrong answer?
- Principle of Least Privilege: Does the LLM really follow the PoLP (Trivially, only for Access Control tasks...)

Cybersecurity

Ransomware Attacks



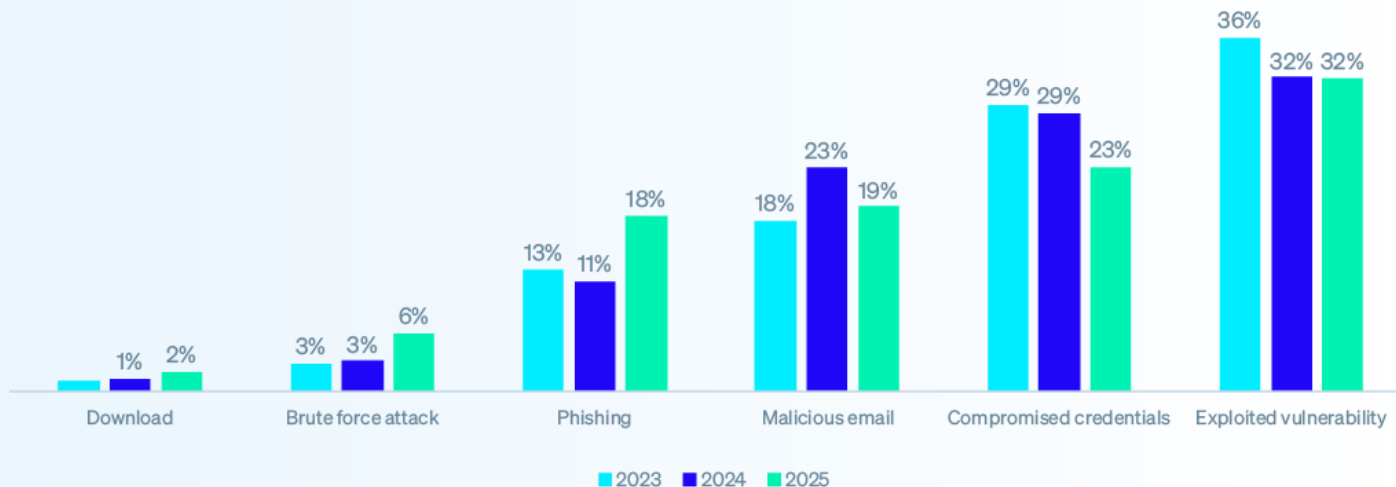
How ransomware works

Source: <https://www.akamai.com/glossary/what-is-ransomware>



Ransomware Attacks

Technical root cause of ransomware attacks 2023–2025



Do you know the root cause of the ransomware attack your organization experienced in the last year? Yes. n=3,400 (2025), 2,974 (2024), 1,974 (2023).

SOPHOS

Source: <https://www.sophos.com/en-us/blog/the-state-of-ransomware-2025>

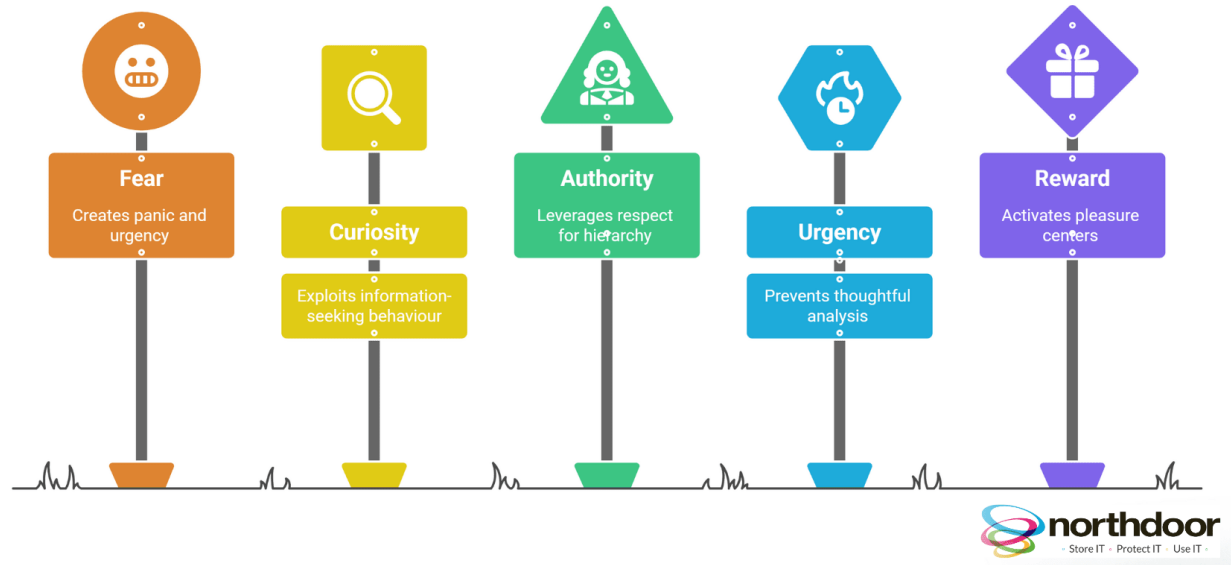
Human Factor in Cybersecurity



Source: <https://threatcop.com/blog/top-5-cyber-attacks-and-security-breaches-due-to-human-error/>

Human Factor in Cybersecurity

Exploiting Emotions: Attacker's Psychological Toolkit



Source: <https://www.northdoor.co.uk/insight/blog/human-factors-in-cyber-security/>

Thank you for your attention!

